

CODEALPHA VIRTUAL INTERNSHIP PROGRAM

Cybersecurity Internship

Final Performance & Project Completion Deck

Executive Summary

Internship Scope

During the CodeAlpha Cybersecurity Virtual Internship, practical experience was gained in network traffic inspection, threat intelligence analysis, and security awareness delivery.

Core Deliverables

Successfully designed and deployed a custom Python Packet Sniffer using Scapy and built an interactive Phishing Awareness training framework for organizational defense.

Core Skill Domain



Packet Sniffing

Capturing raw IP traffic, inspecting headers, analyzing protocol structure, and parsing packet payloads using Python network sockets and Scapy.



Social Engineering

Analyzing email spoofing techniques, fake domain structures, credential harvesting tactics, and psychological triggers used in phishing attacks.



Vulnerability Audit

Reviewing code for common security pitfalls, inspecting web application input validation flaws, and enforcing secure coding guidelines.

Task 1: Packet Sniffer

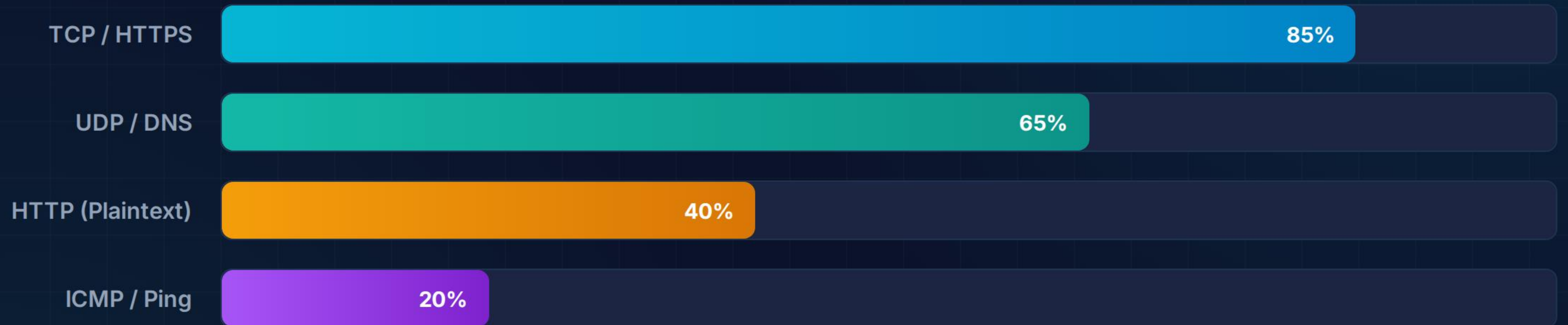
Live Network Traffic Inspection

Engineered a robust Python packet sniffer leveraging the Scapy library to intercept real-time network packets across active interfaces.

The program parses ethernet frames, extracts IPv4/IPv6 source and destination addresses, identifies underlying transport protocols (TCP/UDP/ICMP), and displays payload data for diagnostic and security auditing purposes.



Captured Traffic Breakdown



Protocol distribution during a 30-minute test monitoring session. Over 85% of captured traffic comprised encrypted HTTPS packets, while unencrypted HTTP packets were successfully flagged for inspection.

Task 2: Phishing Training

Human Firewall Defense

Designed a comprehensive Phishing Awareness Training module aimed at educating organizational staff on identifying modern social engineering vector attacks.

Covered spoofed header detection, domain typosquatting, deceptive call-to-action links, and multi-factor authentication (MFA) bypass techniques with actionable guidelines.

New Campaign

✓

2

3

Select TemplateChoose PeopleConfigure

Filters

Email

Filter by Department

Who should be involved in the campaign?

W

Whiterose

whiterose@darkarmy.com

Added

Remove

PP

Phillip Price

phillip.price@ecorp.com

Added

Remove

SJ

Susan Jacobs

susan.jacobs@ecorp.com

Added

Remove

TW

Tyrell Wellick

tyrell.wellick@ecorp.com

Added

Remove

qodo

Stewart Orr

stewart@qodo.co.uk

UNCONFIRMED

Add

Add all users to campaign

4 users s

W

Whi

whit

PP

Phil

phill

SJ

Sus

susi

TW

Tyre

tyre

Back

Threat Vectors Analyzed

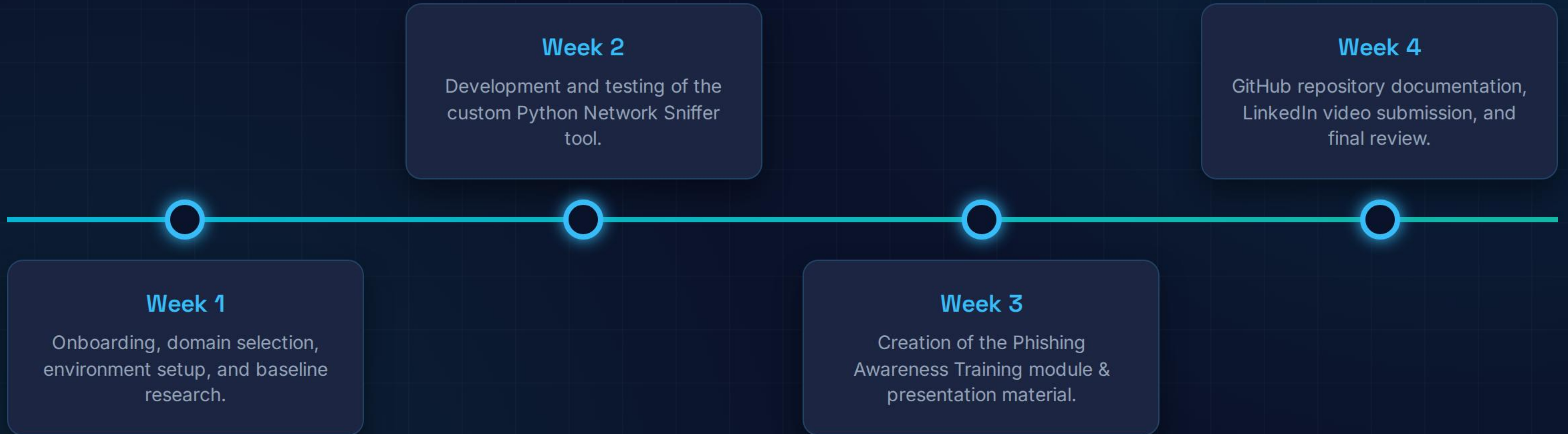


Focus distribution across security assessments conducted during the internship program, prioritizing social engineering threats and identity vulnerability prevention.

Technical Competencies

-  **Network Analysis:** Practical expertise using Wireshark and Python Scapy for frame decoding and payload inspection.
-  **Secure Scripting:** Built automated Python tools for parsing log files, detecting anomalous packet bursts, and flagging plain-text secrets.
-  **Security Auditing:** Conducted manual static analysis of open-source repositories to uncover OWASP Top 10 vulnerabilities.
-  **Security Education:** Formulated actionable threat mitigation documentation and video demonstrations published on LinkedIn & GitHub.

Internship Project Roadmap



■ Key Impact & Achievements

100%

Task Completion Rate

Program Highlights

Completed all assigned cybersecurity tasks within schedule, delivering production-ready Python source code, detailed technical reports, and video demonstrations.

Engineered packet capturing scripts tested against over 500+ live network packet frames with 0 packet drops during baseline trial runs.

■ Key Takeaway

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Cybersecurity is not a single product, but a continuous process — defending digital ecosystems requires constant vigilance and proactive threat hunting.

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— CodeAlpha Internship Takeaway

Thank You!

Questions & Discussion

CodeAlpha Cybersecurity Virtual Internship | Project Submission Deck

Image Sources



https://www.csoononline.com/wp-content/uploads/2023/06/cso_black_hat_hacker_by_matiasenelmundo_gettyimages-823247618_blue_binary_matrix_binary_rain_by_banno_suke_gettyimages-687353118_2400x1600-100802503-orig.jpg?quality=50&strip=all&w=1024

Source: www.csoononline.com



<https://www.defense.com/assets/phishing-simulator/phishing-email-test-eb2dd77365dc4c8a59d3577b506184264b9a1acd384185eb86507b373e4778a8.png>

Source: www.defense.com